THE AUTOMORPHISM COUNT OF A FINITE EXTENSION EQUALS ITS SEPARABLE DEGREE EXACTLY WHEN THE EXTENSION IS NORMAL

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file fields.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: The recall 'Recall that [E : F]_s = |Mor_F(E, F-bar)|' (source lines 2220-2221) is lemma-separable-degree of the same chapter (delivered card L48344 in this pool).

Editorial note. The label lemma-normal-and-automorphisms names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Same chapter: lemma-separable-degree ([E : F]_s = |Mor_F(E, F-bar)|), lemma-characterize-normal (normality iff all F-embeddings into an algebraic closure have the same image)

Source reference. Original reference 09HQ: sources/stacks-fields/pinned-original-source.tex, source lines 2096–2096. Prerequisite pointer invoked at 2230; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let E/F be a finite extension. We have*

$$|Aut(E/F)| \leq [E : F]_s$$

with equality if and only if E is normal over F.

Proof. Choose an algebraic closure $\overline{F}$ of F. Recall that $[E : F]_s = |\operatorname{Mor}_F(E, \overline{F})|$. Pick an element $\sigma_0 \in \operatorname{Mor}_F(E, \overline{F})$. Then the map

$$\operatorname{Aut}(E/F) \longrightarrow \operatorname{Mor}_F(E, \overline{F}), \quad \tau \longmapsto \sigma_0 \circ \tau$$

is injective. Thus the inequality. If equality holds, then every $\sigma \in \operatorname{Mor}_F(E, \overline{F})$ is gotten by precomposing σ_0 by an automorphism. Hence $\sigma(E) = \sigma_0(E)$. Thus E is normal over F by Lemma 09HQ.

Conversely, assume that E/F is normal. Then by Lemma 09HQ we have $\sigma(E) = \sigma_0(E)$ for all $\sigma \in \operatorname{Mor}_F(E, \overline{F})$. Thus we get an automorphism of E over F by setting $\tau = \sigma_0^{-1} \circ \sigma$. Whence the map displayed above is surjective. □

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